

Week 2 module

Page No. _____

Date _____

Q1

find the student roll number
(using linear search)

Problem Statement

Your class teacher has a list of student roll number stored and in the order they registered. Since the list is unsorted, you need to find whether a student of linear search.

```
#include <stdio.h>
int main () {
    int N, i, j, search;
    printf ("\\n Enter ");
    scanf ("%d", &N);
    if (N < 1 || N > 1000) {
        printf ("Invalid N");
    }
    int a[N];
```

```

printf("Enter the roll");
for (i=0; i<N; i++) {
    scanf("%d", &a[i]);
}

```

```

printf("Enter roll no. :");
scanf("%d", &search);

```

```

for (i=0; i<N; i++) {
    if (a[i] == search) {
        printf("Xn Found at index %d", i);
        break;
    }
    return 0;
}

```

2 Search Product ID

problem statement:

An online store keeps product ID sorted in ascending order. A customer wants to know whether a particular product exist in inventory

```
#include <stdio.h>
```

```
int main()
```

```
{
    int N, i, j, first, mid, last, sec;
```

```
    printf("No. of product");
```

```
    scanf("%d", &N);
```

```
    if (N < 1 || N > 100000) {
```

```
        printf("Invalid input");
    }
```

```
    int a[N];
```

```
    printf("Enter sorted product ID");
```

```
    for (i = 0; i < N; i++) {
```

```
        scanf("%d", &a[i]);
    }
```

```
printf("\n Target ID ");  
scanf("%d", &search);
```

```
first = 0;  
last = 0;
```

```
while (first <= last) {  
    mid = (first + last) / 2;
```

```
    if (a[mid] == search) {  
        printf("\n present ");  
        break;
```

```
    } else {
```

```
        if (a[mid] < search) {  
            first = mid + 1;
```

```
        } else {
```

```
            last = mid - 1;
```

```
        }
```

```
    }  
    if (first > last) {
```

```
        printf("\n Not found");  
    }
```

```
    return 0;
```

```
}
```

3 Arrange Exam scores

A teacher wants to arrange student exam scores in ascending order to prepare the result sheet.

```
#include <stdio.h>
int main () {
    int N, i, j, t;
    printf ("Enter number of student ");
    scanf ("%d", &N);

    int a[N];
    printf ("Enter score ");
    for (i=0; i<N; i++) {
        scanf ("%d", &a[i]);
    }

    for (i=0; i<N-1; i++) {
        for (j=0; j<N-1; j++) {
            if (a[j] > a[j+1]) {
                t = a[j];
                a[j] = a[j+1];
                a[j+1] = t;
            }
        }
    }
}
```

```

    }
    }
    }
    printf("\n sorted array")
    for (i=0; i<N; i++){
        printf("%d", a[i]);
    }
    return 0;
}

```

4 Berang Library book Member

A librarian receives book serial number one by one & wants to place each new book in the correct position

```
#include <stdio.h>
int main() {
    int N, i, j, x;
    printf("Enter no of book ");
    scanf("%d", &N);
```

```
int o[N];
    printf("Enter scores ");
    for (i=0; i<N; i++) {
        scanf("%d", &o[i]);
```

```
    for (j=1; j<N; j++) {
        j = o[j];
        j = j-1;
```

```

while (j >= 0 && a[j] > x) {
    a[j+1] = a[j];
    j = j - 1;
}
a[j+1] = x;
}

```

```

printf("sorted books");
for (i = 0; i < N; i++) {
    printf("%d", a[i]);
}
return 0;
}

```

5 Rank Players by score:

A gaming club wants to rank player based on scores from lowest to highest.

```
#include <stdio.h>
int main() {
    int N, i, j, x, min;
    printf("Enter number of player");
    scanf("%d", &N);
    int a[N];
    printf("Enter player scores:");
    for (i=0; i<N; i++){
        scanf("%d", &a[i]);
    }

    for (i=0; i<N-1; i++){
        min = i;
        for (j=i+1; j<N; j++){
            if (a[j] < a[min]) {
                min = j;
            }
        }
    }
}
```

```
    x = a[min];  
    a[min] = a[i];  
    a[i] = x;  
}
```

```
    x = a[min];  
    a[min] = a[i];  
    a[i] = x;  
}
```

```
printf("\n sorted ");  
for (i = 0; i < N; i++) {  
    printf("%d", a[i]);  
}  
return 0;
```

Problem Statement 1 Solution:

```
C Main.c x + Console I/O AI Agent

1 //Find student roll number (Linear Search)
2
3 #include<stdio.h>
4 int main() {
5     int N, i, search;
6     printf("\nEnter number of students: ");
7     scanf("%d", &N);
8
9     if (N < 1 || N > 1000) {
10         printf("Invalid N");
11         return 0;
12     }
13
14     int a[N];
15     printf("Enter the roll numbers: ");
16     for(i = 0; i < N; i++) {
17         scanf("%d", &a[i]);
18     }
19
20     printf("Enter roll number to search: ");
21     scanf("%d", &search);
22
23     for(i = 0; i < N; i++) {
24         if (a[i] == search) {
25             printf("\nFound at index %d", i);
26             return 0;
27         }
28     }
29
30
31     return 0;
32 }
```

Enter number of students: 5
Enter the roll numbers: 101 105 113 110 130
Enter roll number to search: 110

Found at index 3

Problem Statement 2 Solution:

```
C Main.c x + Console I/O AI Agent

1 //Search Product ID(binary Search)
2 #include<stdio.h>
3
4 int main() {
5     int N, i, first, mid, last, search;
6     printf("\nEnter number of products: ");
7     scanf("%d", &N);
8
9     if (N < 1 || N > 100000) {
10         printf("Invalid N");
11         return 0;
12     }
13
14     int a[N];
15     printf("Enter sorted product IDs: ");
16     for (i = 0; i < N; i++) {
17         scanf("%d", &a[i]);
18     }
19
20     printf("Enter target ID: ");
21     scanf("%d", &search);
22
23     first = 0;
24     last = N - 1;
25
26     while (first <= last) {
27         mid = (first + last) / 2;
28
29         if (a[mid] == search) {
30             printf("\nProduct available");
31             return 0;
32         } else if (a[mid] < search) {
33             first = mid + 1;
34         } else {
35             last = mid - 1;
36         }
37     }
38
39     printf("\nProduct not available");
40     return 0;
41 }
```

Enter number of products: 6
Enter sorted product IDs: 10 20 30 40 50 60
Enter target ID: 40

Product available

Problem Statement 3 Solution:

<pre>1 // Arrange Exam score (Bubble Sort) 2 3 #include <stdio.h> 4 5 int main() { 6 int N, i, k, t; 7 printf("\nEnter number of students: "); 8 scanf("%d", &N); 9 10 int a[N]; 11 printf("Enter scores: "); 12 for (i = 0; i < N; i++) { 13 scanf("%d", &a[i]); 14 } 15 16 for (i = 0; i < N - 1; i++) { 17 for (k = 0; k < N - i - 1; k++) { 18 if (a[k] > a[k + 1]) { 19 t = a[k]; 20 a[k] = a[k + 1]; 21 a[k + 1] = t; 22 } 23 } 24 } 25 26 printf("\nSorted array: "); 27 for (i = 0; i < N; i++) { 28 printf("%d ", a[i]); 29 } 30 printf("\n"); 31 return 0; 32 }</pre>	<pre>Enter number of students: 5 Enter scores: 89 45 78 12 99 Sorted array: 12 45 78 89 99</pre>
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Problem Statement 4 Solution:

<pre>1 // Arrange Library book numbers (Insertion Sort) 2 3 #include <stdio.h> 4 5 int main() { 6 int N, i, k, t; 7 8 printf("\nEnter number of books: "); 9 scanf("%d", &N); 10 11 int a[N]; 12 printf("Enter book numbers: "); 13 for (i = 0; i < N; i++) { 14 scanf("%d", &a[i]); 15 } 16 17 for (i = 1; i < N; i++) { 18 t = a[i]; 19 k = i - 1; 20 21 while (k >= 0 && a[k] > t) { 22 a[k + 1] = a[k]; 23 k = k - 1; 24 } 25 26 a[k + 1] = t; 27 } 28 29 printf("Sorted book numbers: "); 30 for (i = 0; i < N; i++) { 31 printf("%d ", a[i]); 32 } 33 printf("\n"); 34 35 return 0; 36 }</pre>	<pre>Enter number of books: 5 Enter book numbers: 55 23 87 11 34 Sorted book numbers: 11 23 34 55 87</pre>
---	--

Problem Statement 5 Solution:

```
C Main.c × +  
1 // Rank players by score(selection sort)  
2  
3 #include <stdio.h>  
4 int main() {  
5     int N, i, k, t, min;  
6     printf("\n Enter number of players:");  
7     scanf("%d", &N);  
8     int a[N];  
9     printf("Enter player scores:");  
10    for(i=0; i<N; i++) {  
11        scanf("%d", &a[i]);  
12    }  
13    for(i=0; i<N-1; i++) {  
14        min=i;  
15        for(k=i+1; k<N; k++) {  
16            if (a[i]<a[k]) {  
17                min=k;  
18            }  
19        }  
20    }  
21    t=a[min];  
22    a[min]=a[i];  
23    a[i]=t;  
24 }  
25 printf(" \n sorted scores: ");  
26 for (i=0; i<N; i++) {  
27     printf(" %d ", a[i]);  
28 }  
29 printf(" \n ");  
30 return 0;  
31 }  
32 }
```

Enter number of players:5
Enter player scores:70 20 90 10 40
sorted scores: 70 20 90 10 40